Block-aware π weights on labeled Hugging Face datasets

CFPerm prototype

Analyst-defined feature-blocks and Stage-1 consensus π guide Stage-2 training: π-weighted RF (column replicates so `max_features` samples $p_j \propto \pi_m/|B_m|$), a linear opinion pool $\sum_m \pi_m \hat{e}_m$, and $X{+}Z$ stacking. Y is the dataset label; W is the batch/domain used for FSDS.

1 Overview dashboard

2 Chronoberg (temporal text, VAD blocks)

Temporal batch $W = 1750$ vs. 1950 on `spaul25/Chronoberg`; modalities hashed text / valence / arousal / dominance.

3 Adult census – income label – sex as batch

Dataset: `scikit-learn/adult-census-income`. Label Y: income >50K. W = sex (Female vs Male); relationship dropped (Husband/Wife leak). $n_0 = 800$, $n_1 = 800$.

4 Yelp vs Amazon – sentiment label – marketplace as batch

Dataset: `fancyzhx/yelp_polarity vs fancyzhx/amazon_polarity`. Label Y: binary sentiment. W = marketplace (Yelp vs Amazon); reviews truncated to 50 tokens. $n_0 = 800$, $n_1 = 800$.

5 MultiNLI – entailment label – fiction vs telephone

Dataset: `nyu-mll/multi_nli validation_matched`. Label Y: entailment vs rest. W = genre (fiction vs telephone). $n_0 = 800$, $n_1 = 800$.

6 Cross-board notes

- Trees ignore monotone column scaling; π enters sklearn RF through column multiplicity.
- Random-subspace BAWF is a different estimator class (not shown on these boards).
- Adaptive group-logit can saturate on mean-shift inject; the RF analogue is the headline.
- Observational Hugging Face shifts are often diffuse across blocks; the GT inject rows isolate whether π guides training when the shift *is* concentrated.

Table 1: Chronoberg 1750 vs. 1950: modality-specific gap shares and clever-covariate FSDS.

Decomposition	text	valence	arousal	dominance
RF VIMP	0.570	0.172	0.102	0.156
block AUC	0.303	0.267	0.167	0.262
block MMD	0.083	0.619	0.037	0.261
LOMO drop	0.292	0.243	0.150	0.315
consensus π	0.311	0.307	0.131	0.250

Table 2: Human blocks $\rightarrow$ Stage-1 π $\rightarrow$ Stage-2. π-weighted RF replicates columns so sklearn `max_features` samples $p_j \propto \pi_m/|B_m|$.

Setting	raw RF	π-RF	pool	$X+Z$	$\Delta_{\pi-\mathrm{RF}}$	π-RF on GT
inject valence (α=0.70)	0.826	0.916	0.837	0.892	+0.091	0.482
text$\perp$ + inject valence	0.855	0.941	0.865	0.911	+0.085	0.572
synthetic GT=valence	0.880	0.897	0.901	0.902	+0.016	0.828

Table 3: Adult census – income label – sex as batch: Stage-1 block shares. Y=income >50K; W = sex (Female vs Male); relationship dropped (Husband/Wife leak).

Decomposition	demography	work	hours	capital
RF VIMP	0.514	0.306	0.128	0.053
block AUC	0.366	0.341	0.218	0.075
block MMD	0.576	0.164	0.261	0.000
LOMO drop	0.499	0.373	0.127	0.000
consensus π	0.480	0.290	0.181	0.049

Table 4: Adult census – income label – sex as batch: Stage-2 π-guided training (5-fold CV AUC).

Setting	raw RF	π-RF	pool	$X+Z$	$\Delta_{\pi-\mathrm{RF}}$	π-RF on GT
observational	0.855	0.849	0.845	0.857	-0.006	—
inject hours (α=0.40)	0.950	0.987	0.988	0.987	+0.037	0.947
demography$\perp$ + inject hours	0.945	0.987	0.986	0.983	+0.042	0.984

Table 5: Yelp vs Amazon – sentiment label – marketplace as batch: Stage-1 block shares. Y=binary sentiment; W = marketplace (Yelp vs Amazon); reviews truncated to 50 tokens.

Decomposition	text	style	polarity
RF VIMP	0.767	0.151	0.082
block AUC	0.519	0.389	0.092
block MMD	0.435	0.429	0.136
LOMO drop	0.681	0.319	0.000
consensus π	0.575	0.314	0.111

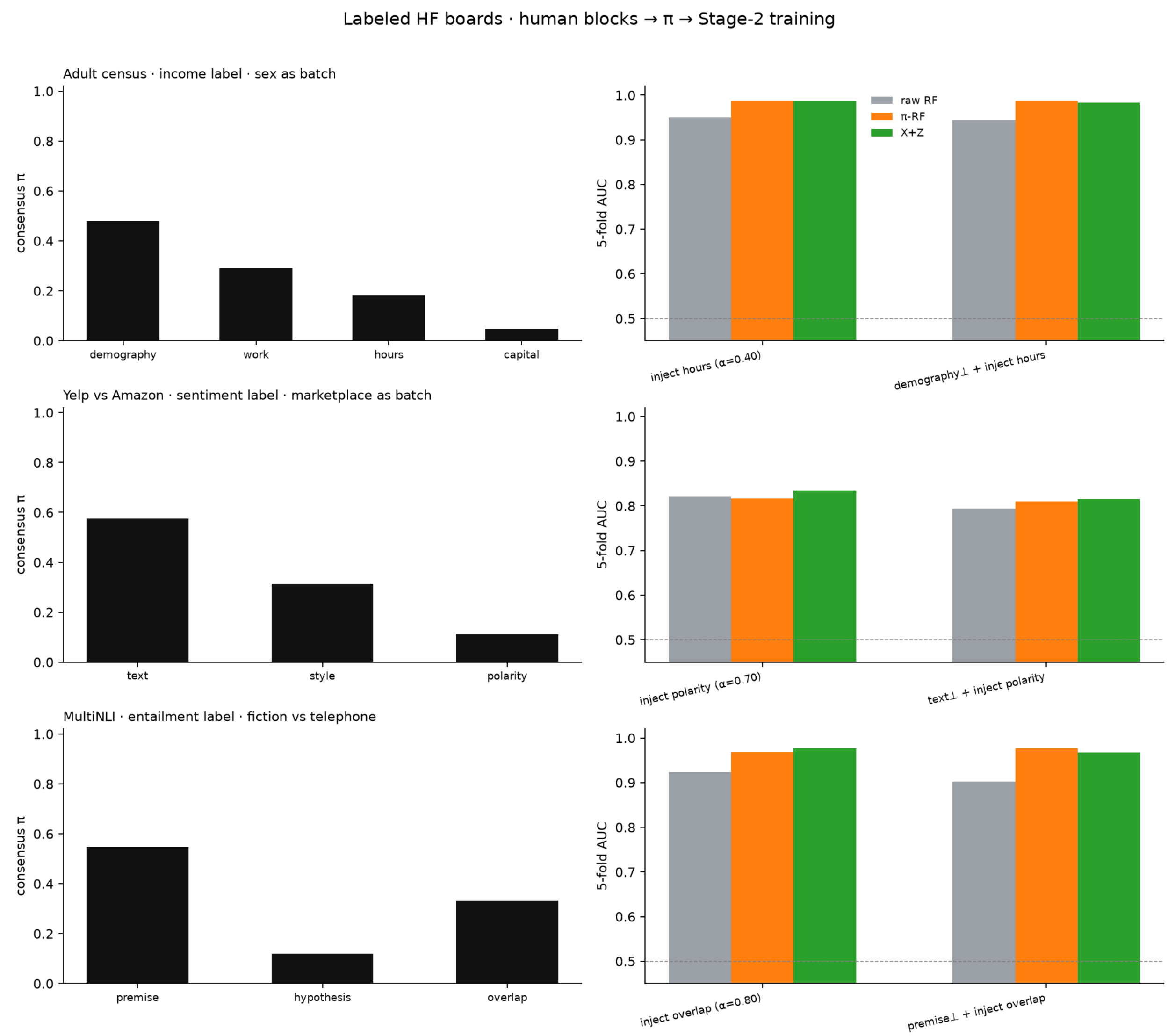
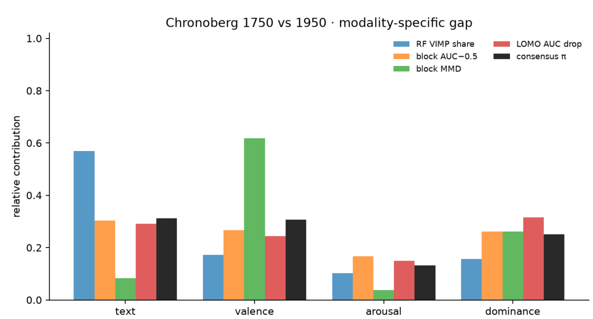
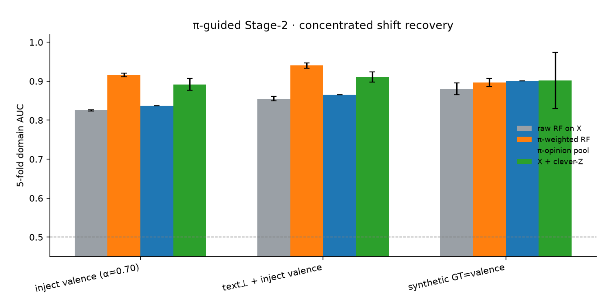


Figure 1: Each row is one labeled Hugging Face board. Left: Stage-1 consensus π over human blocks. Right: 5-fold domain AUC on concentrated inject-GT (raw RF vs π-weighted RF vs $X+Z$).

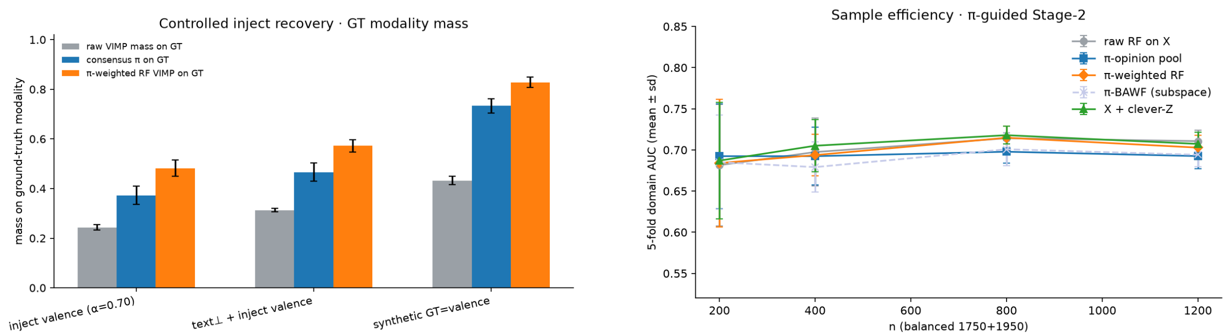


(a) Stage-1 gap shares.



(b) π-guided Stage-2 GT board.

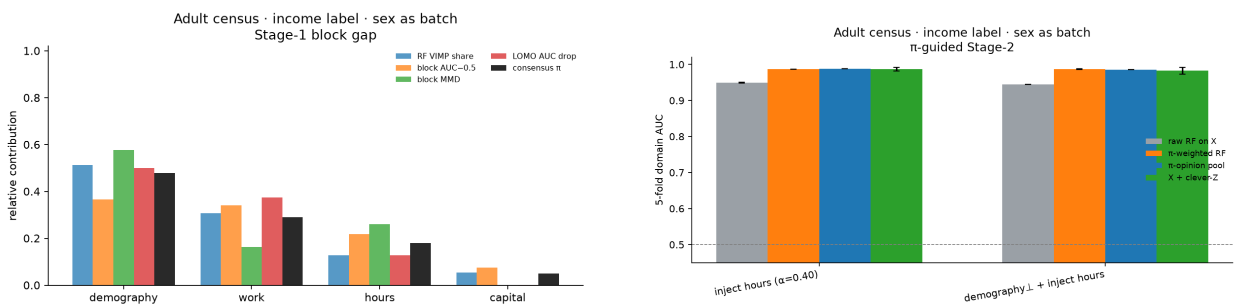
Figure 2: Chronoberg 1750 vs. 1950. Observational shift is diffuse; wins are on concentrated valence inject.



(a) GT mass: raw VIMP vs π vs π-RF.

(b) Sample efficiency on the observational split.

Figure 3: Chronoberg localization and n-scaling.



(a) Stage-1 block gap.

(b) Stage-2 domain AUC.

Figure 4: Adult census – income label – sex as batch: human blocks and π-guided training. Observational rows can be diffuse; inject-GT is the concentrated-shift check.

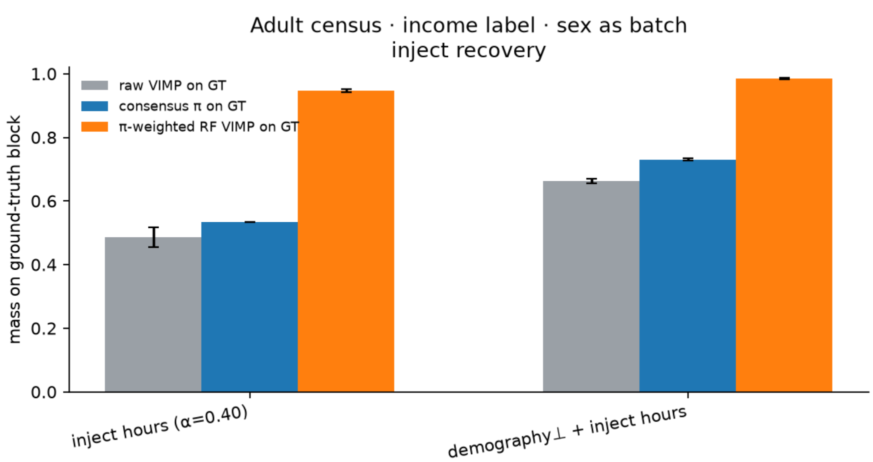
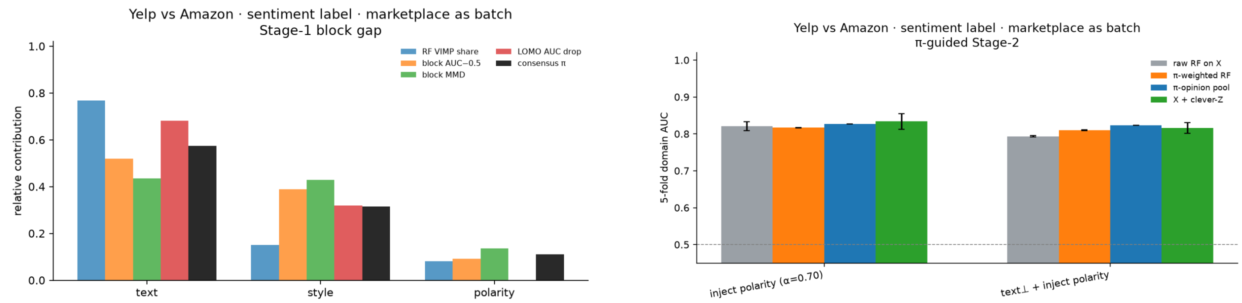


Figure 5: Adult census – income label – sex as batch: mass on the ground-truth block after inject. π-weighted RF should concentrate relative to raw VIMP.



(a) Stage-1 block gap.

(b) Stage-2 domain AUC.

Figure 6: Yelp vs Amazon – sentiment label – marketplace as batch: human blocks and π-guided training. Observational rows can be diffuse; inject-GT is the concentrated-shift check.

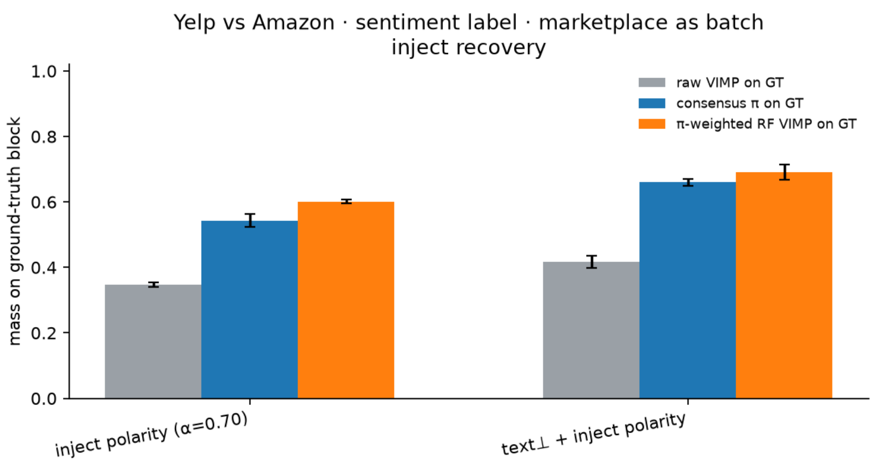


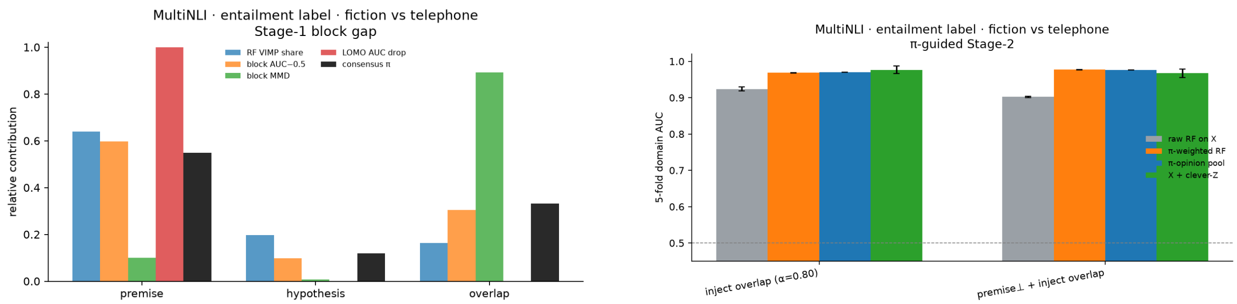
Figure 7: Yelp vs Amazon – sentiment label – marketplace as batch: mass on the ground-truth block after inject. π-weighted RF should concentrate relative to raw VIMP.

Table 6: Yelp vs Amazon – sentiment label – marketplace as batch: Stage-2 π-guided training (5-fold CV AUC).

Setting	raw RF	π-RF	pool	X+Z	$\Delta_{\pi-\mathrm{RF}}$	π-RF on GT
observational	0.728	0.725	0.729	0.727	-0.003	—
inject polarity (α=0.70)	0.821	0.817	0.827	0.834	-0.004	0.602
text⊥ + inject polarity	0.794	0.810	0.823	0.816	+0.016	0.691

Table 7: MultiNLI – entailment label – fiction vs telephone: Stage-1 block shares. Y=entailment vs rest; W = genre (fiction vs telephone).

Decomposition	premise	hypothesis	overlap
RF VIMP	0.640	0.196	0.163
block AUC	0.597	0.099	0.304
block MMD	0.101	0.008	0.891
LOMO drop	1.000	0.000	0.000
consensus π	0.549	0.119	0.332



(a) Stage-1 block gap.

(b) Stage-2 domain AUC.

Figure 8: MultiNLI – entailment label – fiction vs telephone: human blocks and π-guided training. Observational rows can be diffuse; inject-GT is the concentrated-shift check.

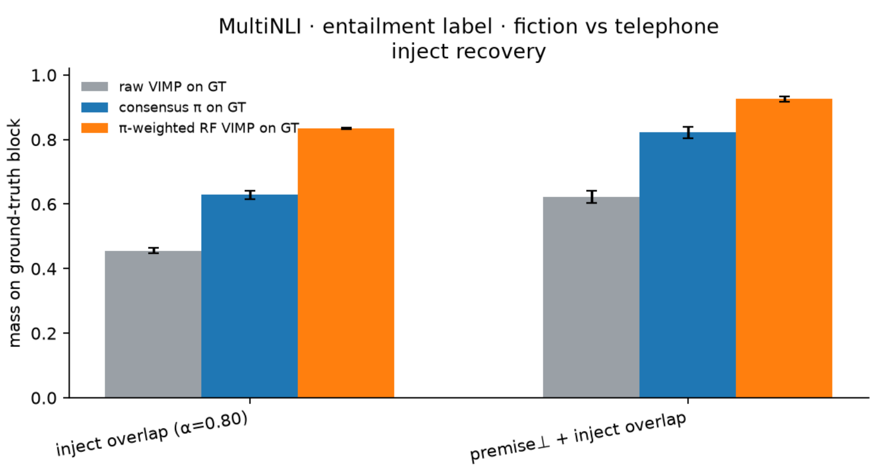


Figure 9: MultiNLI – entailment label – fiction vs telephone: mass on the ground-truth block after inject. π-weighted RF should concentrate relative to raw VIMP.

Table 8: MultiNLI – entailment label – fiction vs telephone: Stage-2 π-guided training (5-fold CV AUC).

Setting	raw RF	π-RF	pool	$X+Z$	$\Delta_{\pi-\mathrm{RF}}$	π-RF on GT
observational	0.836	0.818	0.848	0.875	-0.018	—
inject overlap (α=0.80)	0.924	0.969	0.971	0.977	+0.045	0.836
premise⊥ + inject overlap	0.902	0.977	0.977	0.967	+0.075	0.926